

World Models for Embodied AI: Manipulation, Locomotion and Navigation

SEMINAR REPORT

Submitted by

Navaneeth K P

SNM23CA013

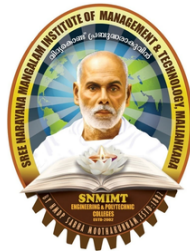
to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree of
Bachelor of Technology In
Department of Computer Science and Engineering (Artificial Intelligence)

Under the guidance of

Ms. Surya Jayasingh P

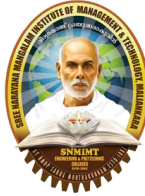


Department of Computer Science and Engineering (Artificial Intelligence)

SREE NARAYANA MANGALAM INSTITUTE OF
MANAGEMENT AND TECHNOLOGY
MALIANKARA

July 2026

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE)
SREE NARAYANA MANGALAM INSTITUTE OF MANAGEMENT &
TECHNOLOGY, MALIANKARA**



CERTIFICATE

*This is to certify that the seminar report entitled **World Models for Embodied AI: Manipulation, Locomotion and Navigation** submitted by Navaneeth K P - SNM23CA013, to the Department of Computer Science and Engineering (Artificial Intelligence), SNM Institute of Management and Technology, Maliankara, in partial fulfilment of the requirement for the award of B.Tech Degree in Computer Science and Engineering (Artificial Intelligence) is a bonafide record of the work carried out by him.*

Guide

Ms. Surya Jayasingh P
Asst. Professor,
Dept. of CSE (AI)

Head of the Department

Ms. Lakshmi Anand
Asst. Professor,
Dept. of CSE (AI)

Coordinator

Ms. Surya T S
Asst. Professor,
Dept. Of CSE (AI)

Declaration

I undersigned hereby declare that the seminar report **World Models for Embodied AI: Manipulation, Locomotion and Navigation**, submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of **Ms. Surya Jayasingh P**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Maliankara

Navaneeth K P

Date:

SNM23CA013

Acknowledgement

Firstly, I thank God, the Almighty, with whose blessings, this seminar has been successfully completed.

I wish to give deep sense of acknowledgement to our Principal **Dr. Atmaram P**, for his constant encouragement and valuable advice throughout the course.

I am extremely grateful to **Ms. Lakshmi Anand**, Head of the department of Computer Science and Engineering (Artificial Intelligence), for the valuable guidance through this humble endeavor. The highly enterprising attitude, patience for listening to my doubts, timely suggestions and guidance made my seminar a so-called reality in stipulated time. I would also like to thank our seminar coordinator **Ms. Surya T S**, Assistant Professor, Department of Computer Science and Engineering (Artificial Intelligence) and my seminar guide **Ms. Surya Jayasingh P**, Assistant Professor, Department of Computer Science and Engineering (Artificial Intelligence) for their gracious of encouragement and the valuable assistance.

I express my sincere gratitude to all the faculty members of the department of Computer Science and Engineering (Artificial Intelligence) for their co-operation and support to the seminar.

I also express my sincere gratitude to all my friends and my parents for their co-operation and constant inspiration.

Navaneeth K P

Abstract

Embodied artificial intelligence requires agents that can perceive, predict and act in the physical world rather than merely map observations to actions. World models—internal representations that simulate how an environment evolves—have emerged as a unifying mechanism for this capability, yet their architectures diverge sharply across embodied domains. This seminar presents a comparative study of four recent IEEE works that build world models for distinct embodied tasks, spanning dexterous manipulation, legged locomotion and object goal navigation.

DexSim2Real², the base paper, constructs an explicit world model by actively interacting with an unseen articulated object, reconstructing a digital twin of it in a physics simulator through 3D AIGC and foundation vision models, and planning long-horizon trajectories using sampling-based Model Predictive Control (MPC), with Eigengrasp employed to reduce the high-dimensional action space of dexterous hands. Gaussian Action Field (GAF) adopts an implicit-geometric route, extending 3D Gaussian Splatting with learnable motion attributes to form a 4D spatiotemporal field from which future scene states and end-effector actions are directly inferred and then refined by an action-vision-aligned denoising module. World Model-based Perception (WMP) applies a latent Recurrent State-Space Model (RSSM) to visual legged locomotion, replacing the conventional teacher-student privileged-learning pipeline with a single-stage framework whose recurrent state compresses depth and proprioceptive history into a control-ready representation that transfers from simulation to hardware across gaps, climbs, tilts and crawls. WMNav pushes the abstraction furthest by treating a Vision-Language Model (VLM) itself as the world model for zero-shot object goal navigation, predicting the likelihood of a target’s presence into an online Curiosity Value Map and decomposing the goal into subtasks to suppress hallucination.

Comparing these systems along the axes of representation (explicit physics model, 4D Gaussian scene field, latent recurrent state, semantic-predictive reasoning), supervision cost, generalization to unseen objects and environments, and real-world deployability

reveals a consistent trade-off between physical fidelity and semantic breadth. The study concludes that hybrid world models, combining geometric precision with commonsense reasoning, represent the most promising direction for the next generation of embodied agents.

Keywords: Embodied AI; World Models; Model-Based Reinforcement Learning (MBRL); Dexterous Manipulation; Sim2Real Transfer; 3D Gaussian Splatting (3DGS); Legged Locomotion; Object Goal Navigation; Vision-Language Models (VLM); Model Predictive Control (MPC).

Contents

Acknowledgement	I
Abstract	II
List of Figures	IV
List of Tables	V
List of Abbreviations	VI
1 Introduction	1
1.1 General Background	4
1.2 Scope	4
1.3 Advantages	5
2 Literature Survey	6
2.1 DexSim2Real ² : Building Explicit World Model for Precise Articulated Ob- ject Dexterous Manipulation	7
2.1.1 System Architecture and Methodology	7
2.2 GAF: Gaussian Action Field as a 4D Representation for Dynamic World Modeling in Robotic Manipulation	9
2.2.1 System Architecture and Methodology	9
2.3 World Model-based Perception for Visual Legged Locomotion (WMP) . . .	11
2.3.1 System Architecture and Methodology	11
2.4 WMNav: Integrating Vision-Language Models into World Models for Ob- ject Goal Navigation	13
2.4.1 System Architecture and Methodology	13
3 Discussion & Comparison	15
4 Challenges	23
5 Conclusion	25
References	27

List of Figures

2.0.1 Taxonomy of World-Model Approaches in Embodied AI	6
2.1.1 [1] Overview of the DexSim2Real ² framework	7
2.2.1 [2] GAF manipulation pipeline with action-vision-aligned denoising	9
2.3.1 [3] Illustration of the WMP framework with RSSM world model	12
2.4.1 [4] The WMNav framework with VLM-based world model and navigation policy	14

List of Tables

3.1 Comparison of World-Model Approaches for Embodied AI	16
--	----

List of Abbreviations

AIGC	Artificial Intelligence Generated Content
AMP	Adversarial Motion Priors
CEM	Cross-Entropy Method
CNN	Convolutional Neural Network
DOF	Degrees of Freedom
GAF	Gaussian Action Field
GRU	Gated Recurrent Unit
3DGS	3D Gaussian Splatting
ICP	Iterative Closest Point
iCEM	Improved Cross-Entropy Method
IL	Imitation Learning
LLM	Large Language Model
MBRL	Model-Based Reinforcement Learning
MLP	Multi-Layer Perceptron
MPC	Model Predictive Control
MPPI	Model Predictive Path Integral Control
PCA	Principal Component Analysis
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RGBD	Red Green Blue plus Depth
RL	Reinforcement Learning
RSSM	Recurrent State-Space Model
SPL	Success weighted by Path Length
SR	Success Rate
URDF	Unified Robot Description Format
ViT	Vision Transformer
VLM	Vision-Language Model
WMP	World Model-based Perception
ZSON	Zero-Shot Object Navigation

Chapter 1

Introduction

Embodied Artificial Intelligence (AI) concerns intelligent agents—most prominently robots—that must perceive their surroundings, reason about the physical world, and execute actions to accomplish goals. Unlike disembodied AI systems that operate purely on digital data, embodied agents are embedded in dynamic, partially observable, and physically constrained environments. A robot that manipulates a cabinet, a quadruped that climbs over an obstacle, and a domestic robot that searches for a bed in an unfamiliar house all face the same fundamental difficulty: a single observation rarely reveals everything the agent needs to know, and every action has physical consequences that unfold over time. Purely reactive policies, which map current perception directly to actions, often struggle in such settings because they cannot anticipate the outcome of an action before committing to it.

Humans and animals solve this problem with remarkable elegance. Cognitive science suggests that biological agents build a **mental model** of the world—an internal representation that captures how the environment behaves and how it will respond to actions. Using motor imagery and mental simulation, humans can “imagine” the consequences of a movement before executing it, and they actively interact with objects to gather the information required to refine this model, a process known as **interactive perception**. Translating this natural ability to machines has given rise to the concept of the **world model**: a computational representation of environment dynamics that allows an agent to predict future states, simulate candidate actions internally, and plan behavior with foresight rather than reacting blindly.

World models have a long history in optimal control and Model-Based Reinforcement Learning (MBRL), where a learned transition model of the environment is used to improve

sample efficiency and enable planning. Modern instantiations span a wide spectrum. At one end are **explicit physics-based world models**, in which the environment is reconstructed as an interactable digital twin inside a physics simulator; these models embed strong physical priors and generalize naturally to unseen actions. At the other end are **learned latent world models**, such as the Recurrent State-Space Model (RSSM) popularized by the Dreamer family, which compress high-dimensional sensory streams into compact latent states that support future prediction. Between these extremes lie structured scene-level representations, such as dynamic 3D Gaussian fields that attach motion attributes to explicit geometry, and, most recently, **foundation-model-driven world models**, in which Large Language Models (LLMs) and Vision-Language Models (VLMs) serve as commonsense predictors of how a scene is likely to unfold.

This seminar examines how these different families of world models are applied across the three core competencies of embodied AI. In the domain of **manipulation**, the base paper DexSim2Real² [1] constructs an explicit world model of unseen articulated objects, such as drawers, cabinets, laptops, and lamps, through one-step interactive perception followed by 3D AI-Generated Content (AIGC)-based digital twin reconstruction; sampling-based Model Predictive Control (MPC) then plans precise, goal-conditioned trajectories for suction grippers, two-finger grippers, and multi-finger dexterous hands without requiring demonstrations or reinforcement learning. Complementing this, the Gaussian Action Field (GAF) [2] adopts a Vision-to-4D-to-Action (V-4D-A) paradigm: it extends 3D Gaussian Splatting (3DGS) with learnable motion attributes so that a single feed-forward network reconstructs the current scene, predicts the future scene, and extracts an initial action from the predicted Gaussian motion, which is subsequently refined by an action-vision-aligned diffusion model.

In the domain of **locomotion**, World Model-based Perception (WMP) [3] addresses visual legged locomotion over challenging terrains such as gaps, climbs, tilts, and crawl spaces. Instead of the conventional privileged teacher–student framework, WMP trains an RSSM world model entirely in simulation to predict future depth images and proprioception, and feeds the recurrent latent state of this model to a Proximal Policy Optimization

(PPO) policy. Although trained purely on simulated data, the world model produces accurate predictions of real-world trajectories, enabling a Unitree A1 quadruped to traverse a gap of 85 cm (about $2.1\times$ its body length) and a climb of 55 cm (about $2.2\times$ its height). In the domain of **navigation**, WMNav [4] integrates VLMs into a modular world model for Zero-Shot Object Navigation (ZSON): a PredictVLM estimates the likelihood that the target object lies in each direction, an online Curiosity Value Map accumulates these predictions as spatial memory, and subtask decomposition with a two-stage action proposer converts the predictions into efficient navigation actions, achieving state-of-the-art success rates on the HM3D and MP3D benchmarks.

Viewed together, the four systems reveal a common architectural philosophy—predict before acting—realized through very different design choices. DexSim2Real² places its knowledge in an explicit simulator and searches for trajectories; GAF embeds dynamics directly into an explicit scene representation and reads actions out of predicted motion; WMP compresses history into a latent state that a learned policy consumes; and WMNav delegates prediction to the commonsense knowledge stored in pre-trained VLMs. These choices induce distinct trade-offs in data requirements, computational cost, precision, interpretability, and generality, which this report analyzes in depth.

Despite their successes, world-model-based embodied AI still faces open **challenges**. Explicit reconstruction pipelines depend on the fidelity of 3D generative models and can be misled by novel geometry; latent models trained in simulation inherit the sim-to-real gap of their training distribution; VLM-based predictors are susceptible to hallucination; and all approaches must balance model accuracy against the latency constraints of real-time control. Emerging directions such as training world models on mixtures of simulated and real data, incorporating tactile and other sensing modalities, scene-level digital twins for mobile manipulation, and hallucination-resistant prompting strategies aim to address these limitations and are discussed throughout this report.

1.1 General Background

Approaches to embodied decision-making can be broadly categorized into **model-free**, **model-based**, and **hybrid** paradigms. Model-free methods, including most Reinforcement Learning (RL) and Imitation Learning (IL) policies, learn a direct mapping from observations to actions; they are simple to deploy but data-hungry and brittle when the state distribution is complex, as with articulated objects. Model-based methods first acquire a model of the environment—explicit or learned—and then use it for planning or policy optimization, gaining sample efficiency and foresight at the cost of model construction. Hybrid systems combine both: a generalizable model-construction module paired with task-specific planning (as in DexSim2Real²), or a learned world model whose latent state feeds a model-free policy (as in WMP). Architecturally, world models exploit components such as Convolutional Neural Networks (CNNs) for visual encoding, Gated Recurrent Units (GRUs) for temporal dynamics, Vision Transformers (ViTs) for geometry-aware feature extraction, and foundation VLMs for semantic prediction, allowing embodied agents to perceive, remember, and predict in real time.

1.2 Scope

The scope of this seminar is to explore and evaluate world-model-driven advancements across the principal domains of embodied AI. It draws from four core research studies focusing on: paper [1] the construction of explicit digital-twin world models of articulated objects through interactive perception and 3D AIGC for precise dexterous manipulation; paper [2] 4D spatiotemporal world modeling with Gaussian Action Fields that unify scene reconstruction, future prediction, and action generation; paper [3] latent RSSM world models as perceptual backbones for vision-based legged locomotion with sim-to-real transfer; and paper [4] VLM-powered predictive world models with curiosity-based spatial memory for zero-shot object goal navigation. The seminar further discusses the evaluation methodologies of these systems (manipulation error, success rate, reconstruction quality, traversal difficulty, SR/SPL navigation metrics), their experimental platforms, and the practical challenges of real-world deployment.

1.3 Advantages

The integration of world models into embodied AI offers numerous advantages that transform reactive robotic systems into predictive, anticipatory agents:

- **Foresight and Planning:** Agents simulate the consequences of candidate actions internally before physical execution, enabling long-horizon planning.
- **Data Efficiency:** Explicit physics priors (DexSim2Real²) and learned dynamics (WMP) drastically reduce the number of real-world interactions required.
- **Demonstration-Free Operation:** Sampling-based MPC over a constructed world model eliminates the need for human demonstrations or reward engineering per task.
- **Precision:** Digital-twin-based planning achieves goal-conditioned manipulation with average errors as low as 7° on real articulated objects.
- **Generalization to Unseen Actions:** An explicit model supports novel strategies never seen during data collection, such as tool-based drawer opening.
- **Robust Sim-to-Real Transfer:** World models trained in simulation predict real-world trajectories accurately, shrinking the sim-to-real gap in locomotion.
- **Reduced Risky Interactions:** Predicting outcomes before acting reduces collisions and costly exploratory failures in navigation.
- **Zero-Shot Semantic Reasoning:** VLM-based world models exploit internet-scale commonsense to navigate to unseen object categories without task-specific training.
- **Unified Perception–Prediction–Action:** Representations such as GAF couple reconstruction, prediction, and action inference in one framework.
- **Cross-Domain Applicability:** The same predictive philosophy applies to manipulation, locomotion, and navigation, indicating a general paradigm for robot control.

Chapter 2

Literature Survey

This chapter reviews four selected research papers under the theme of **World Models for Embodied AI**. Each work instantiates the world-model paradigm in a different embodied domain and with a different representation: an explicit digital-twin world model for precise articulated object manipulation, a 4D Gaussian Action Field for dynamic manipulation scenes, a latent recurrent world model for visual legged locomotion, and a VLM-driven predictive world model for zero-shot object goal navigation. Together, these studies illustrate the evolution of embodied agents from reactive perception–action mappings toward predictive, simulation-capable intelligence.

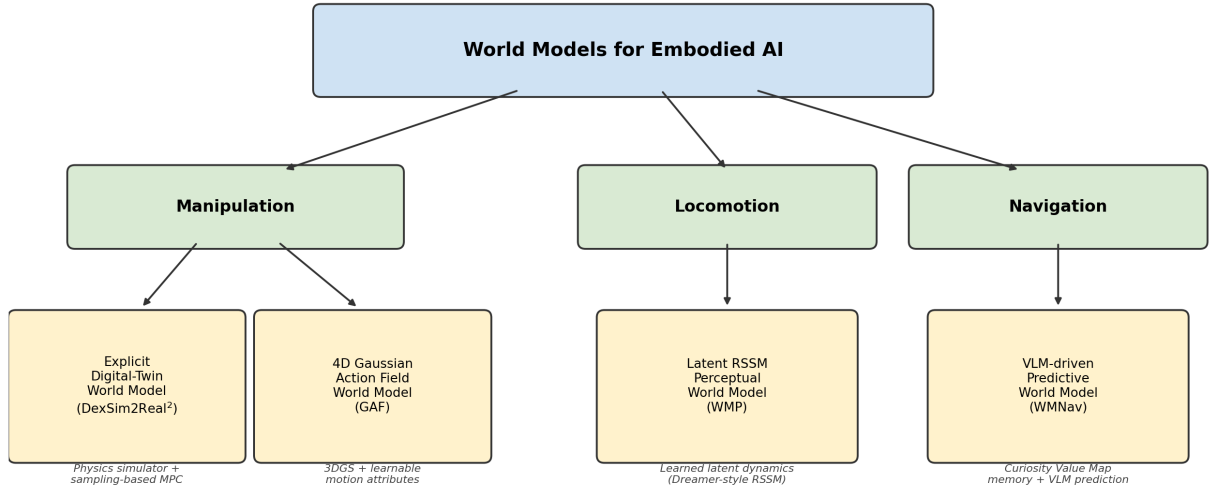


Figure 2.0.1: Taxonomy of World-Model Approaches in Embodied AI

2.1 DexSim2Real²: Building Explicit World Model for Precise Articulated Object Dexterous Manipulation

This paper, the base paper of the seminar, proposes a robot learning framework for precise, goal-conditioned manipulation of unseen articulated objects using suction grippers, two-finger grippers, and multi-finger dexterous hands. Its core idea is to use a physics simulator as the robot’s “mental model”: the robot first interacts with the object to reveal its structure, then constructs an explicit digital twin of the object in simulation, and finally plans long-horizon manipulation trajectories inside this world model using sampling-based Model Predictive Control (MPC), without demonstrations or reinforcement learning.

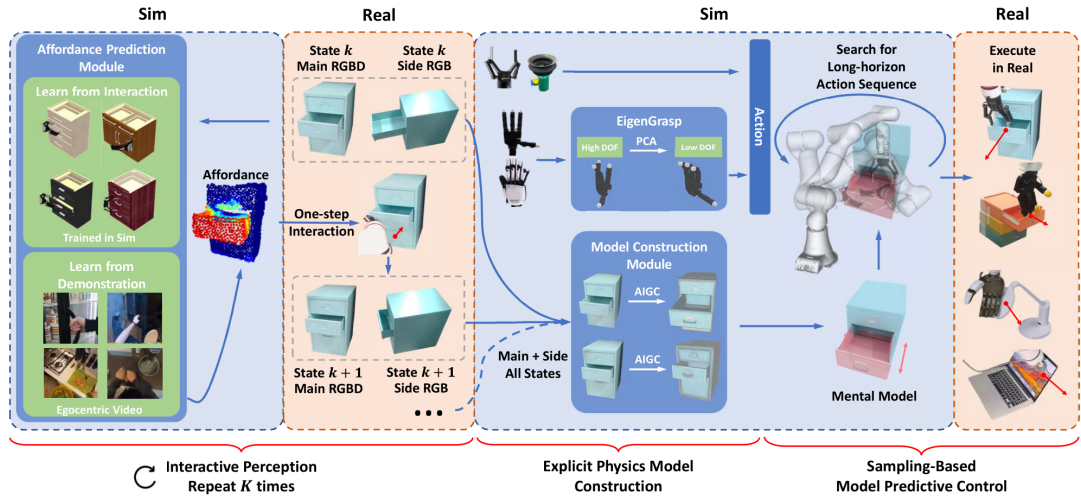


Figure 2.1.1: [1]Overview of the DexSim2Real² framework

2.1.1 System Architecture and Methodology

The proposed framework comprises three sequential modules:

- **Interactive Perception:** An affordance prediction network proposes a one-step interaction that changes the joint state of the articulated object. Affordances are learned either from self-supervised interaction in simulation (Where2Act) or from egocentric human manipulation videos (VRB), where a novel pixel-projection method lifts two 2D post-contact vectors into a single 3D robot motion. For an object with K movable parts, the interaction is repeated K times.

- **Explicit World Model Construction:** From the $K + 1$ frames of observations, a 3D AIGC model (Rodin) generates a high-quality mesh for each state; differentiable rendering estimates pose and scale; foundation vision models (SAMPart3D), mesh connectivity, and robot proprioception jointly segment movable parts; and the kinematic structure (part hierarchy, joint types, axes) is estimated and exported as a URDF digital twin loaded into the SAPIEN simulator after VHACD convex decomposition.
- **Sampling-Based Model Predictive Control:** The improved Cross-Entropy Method (iCEM) searches for a trajectory that drives the object from its initial joint state to the target state, guided by dense rewards (success, approach, contact, distance, and regularization terms).
- **Eigengrasp Dimensionality Reduction:** For dexterous hands (16-DOF Allegro Hand, 12-DOF XHand), Principal Component Analysis over 60,800 grasp postures from DexGraspNet compresses the hand action space; only $m = 2$ eigengrasp coefficients suffice to match the success rate of full-dimensional search while reducing computation time and joint jerk.
- **Evaluation:** Real-world experiments on laptops, cabinets, drawers, and lamps achieve average errors of $7.2^\circ/6.8^\circ$ (suction), $7.2^\circ/7.4^\circ$ (two-finger), and $1.64\text{ cm}/6.87^\circ$ (dexterous hand), significantly outperforming the UMPNet baseline; the explicit model further enables tool-based manipulation of out-of-reach drawers.

By first building the world model of a single object instance and then planning within it, the framework avoids the data-hungry generalization burden of policy networks. The explicit physics prior guarantees that a model constructed from simple one-step interactions supports long-horizon, multi-step, and even unseen tool-using trajectories, marking a distinctive advantage over articulation-flow and policy-based baselines.

2.2 GAF: Gaussian Action Field as a 4D Representation for Dynamic World Modeling in Robotic Manipulation

This research introduces a Vision-to-4D-to-Action (V-4D-A) paradigm in which manipulation actions are reasoned directly from a motion-aware 4D scene representation. The Gaussian Action Field (GAF) extends 3D Gaussian Splatting (3DGS) with a learnable motion attribute per Gaussian, so that a single feed-forward network simultaneously reconstructs the current scene, predicts the future scene, and yields an initial estimate of the robot action from the predicted Gaussian motion.

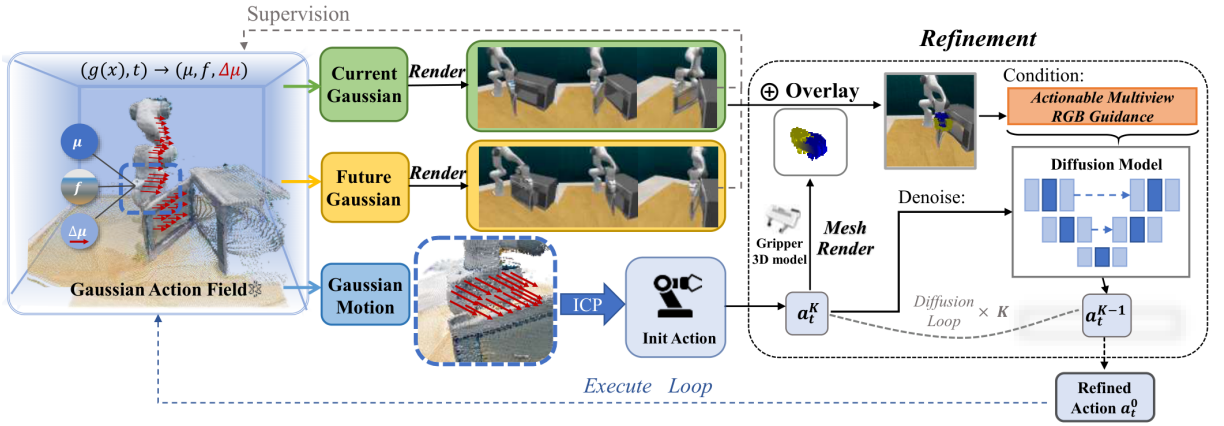


Figure 2.2.1: [2]GAF manipulation pipeline with action-vision-aligned denoising

2.2.1 System Architecture and Methodology

The architecture consists of:

- **Input Encoding:** Two unposed RGB views with intrinsics are tokenized and processed by a shared-weight Vision Transformer (initialized from MAST3R) with cross-view attention; no camera extrinsics or depth are needed for reconstruction.
- **Three Prediction Heads:** A Gaussian Center Head predicts point positions μ , a Gaussian Params Head predicts appearance attributes $f = \{c, \sigma, r, s\}$, and a Motion Prediction Head predicts per-point displacements $\Delta\mu$ over a future interval, together defining current Gaussians $\{\mu, f\}$ and future Gaussians $\{\mu + \Delta\mu, f\}$.

- **Rendering Supervision:** Alpha-blend rendering of current and future Gaussians enables training purely from RGB video frames with LPIPS and MSE losses, avoiding any ground-truth 3D or action labels at this stage.
- **Init Action Computation:** The rigid transformation of the gripper is recovered by Iterative Closest Point (ICP) registration between current and future gripper Gaussians and interpolated into an initial action sequence.
- **Action-Vision-Aligned Denoising:** Candidate gripper poses are mesh-rendered onto the multi-view images to form Actionable Multiview RGB Guidance, conditioning a diffusion model that refines the init action into physically consistent, executable end-effector poses.
- **Evaluation:** On nine RL Bench tasks GAF attains a 60.4% average success rate (+15.7% over Diffusion Policy, +7.3% over Act3D, +10.3% over ManiGaussian) and reconstruction gains of +11.54 dB PSNR, +0.386 SSIM, and -0.557 LPIPS; real-world deployment on a Franka Emika Panda achieves 6–10/10 success across five tasks with 3 diffusion iterations and sub-0.3 s action prediction.

By embedding dynamics directly into an explicit scene representation, GAF closes the mismatch between static scene understanding and dynamic action generation. The predicted future Gaussians are not merely auxiliary supervision, as in prior Gaussian world models, but the very medium from which actions are extracted, producing a tight, interpretable coupling between prediction and control.

2.3 World Model-based Perception for Visual Legged Locomotion (WMP)

This paper presents a simple yet effective end-to-end framework that combines Model-Based Reinforcement Learning (MBRL) world modeling with sim-to-real transfer for vision-based quadrupedal locomotion. Instead of the two-stage privileged teacher–student paradigm, WMP builds a Recurrent State-Space Model (RSSM) of the environment in simulation and learns a policy directly on top of the world model’s recurrent state, thereby avoiding both the information gap between teacher and student and the labor-intensive design of privileged observations.

2.3.1 System Architecture and Methodology

- **Problem Formulation:** Locomotion is cast as a Partially Observable Markov Decision Process (POMDP) whose observation combines proprioception $o_t^p \in \mathbb{R}^{45}$ and a 64×64 egocentric depth image; privileged information (scandots, contact forces, physical parameters) is available only to the critic in simulation.
- **World Model (RSSM):** A GRU-based recurrent model, CNN/MLP encoder, dynamic predictor, and decoder are trained jointly with a reconstruction loss and a KL regularizer; the model runs at a lower frequency than the policy (interval $k = 5$, i.e., 0.1 s) to accommodate the 40 ms on-board latency of depth acquisition and inference.
- **Policy Learning:** An asymmetric actor–critic trained with Proximal Policy Optimization (PPO) receives proprioception plus the stop-gradient recurrent state h_t ; rewards combine clipped velocity tracking with an Adversarial Motion Priors (AMP) style term for natural gaits. Notably, simulator rollouts are used directly, since the world model cannot exceed the simulator’s accuracy.
- **Training Environment:** 4096 Unitree A1 instances are simulated in Isaac Gym on six terrain types (Slope, Stair, Gap, Climb, Tilt, Crawl) with terrain curriculum, domain randomization, and 100 ms perception latency modeling.

2.4 WMNav: Integrating Vision-Language Models into World Models for Object Goal Navigation

This study proposes a fully modular, training-free world-model framework for Zero-Shot Object Navigation (ZSON), in which Vision-Language Models (VLMs) carry out all perception, prediction, planning, and reasoning. Instead of interacting with the environment to discover the target by trial and error, the agent imagines the likely outcome of each direction of travel—exploiting the commonsense knowledge of indoor layouts embedded in VLMs—and stores these predictions in an online memory that steers exploration.

2.4.1 System Architecture and Methodology

- **Panoramic Perception:** At each step the agent rotates through six headings to compose an RGB-D panoramic image, providing perception beyond the limited egocentric field of view.
- **PredictVLM and Curiosity Value Map:** A predictor VLM scores every direction (0–10) for the likelihood of containing the goal; the scores are projected onto a top-down Curiosity Value Map that is merged with the previous map by a pixel-wise minimum, forming the world model’s persistent memory of predicted environmental states.
- **Subtask Decomposition (PlanVLM):** The highest-scoring direction image, together with the previous subtask, is passed to a planner VLM that emits a new intermediate subtask and a goal flag, providing denser feedback and mitigating hallucination by comparing plan against observation.
- **Two-Stage Action Proposer (ReasonVLM):** In the exploration stage, candidate polar-coordinate actions sampled from navigable regions are filtered and annotated on the image for the VLM to select; in the goal-approaching stage, denser unconstrained sampling localizes the detected goal precisely, and stopping is triggered by a distance threshold.

- **Evaluation:** With Gemini VLMs and no task-specific training, pre-built maps, or detectors, WMNav surpasses all zero-shot baselines: +3.2% SR and +3.2% SPL on HM3D and +13.5% SR and +1.1% SPL on MP3D; ablations show the Curiosity Value Map, subtask decomposition, and two-stage proposer each contribute measurably (72.2% SR on HM3D v0.2 with all modules).

WMNav shows that pre-trained VLMs can function as abstract world models for embodied navigation: prediction replaces costly physical exploration, and quantitative curiosity scoring forces rigorous, reliable outputs from the VLM. The design decomposes navigation in a human-like manner—imagine, plan, act—and its performance scales directly with the capability of the underlying VLM.

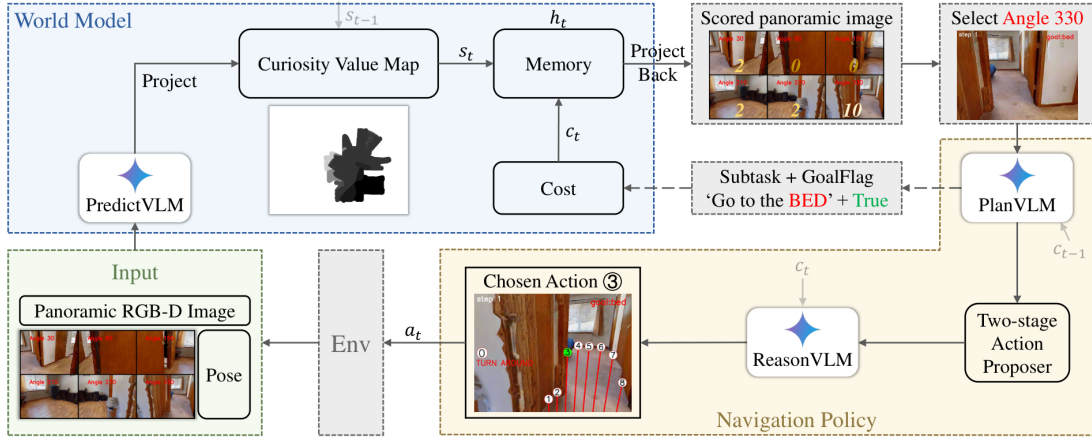


Figure 2.4.1: [4]The WMNav framework with VLM-based world model and navigation policy

Chapter 3

Discussion & Comparison

This chapter presents a detailed comparative analysis of the four reviewed research papers that instantiate world models across the manipulation, locomotion, and navigation domains of embodied AI. These studies collectively demonstrate how predictive internal models—whether explicit physics simulations, motion-augmented Gaussian fields, latent recurrent dynamics, or VLM-based imagination—are transforming robotics from reactive perception–action mappings into anticipatory, plan-before-act intelligence. While each paper shares the common goal of predicting the future state of the world to improve decision-making, they differ fundamentally in representation, learning signals, planning mechanisms, and deployment platforms. The following sections critically evaluate their contributions, advantages, and limitations, offering insights into their comparative strengths and areas for further improvement.

Analytical Focus: Each paper addresses a distinct challenge in world-model-driven embodied AI. Paper [1] tackles **precise, goal-conditioned manipulation of unseen articulated objects**, turning perception into search inside a physics simulator via an explicit digital twin. Paper [2] addresses the **mismatch between static scene understanding and dynamic action generation** by embedding motion into an explicit Gaussian scene representation, so actions are read directly from predicted geometry. Paper [3] targets **perception under partial observability** in legged locomotion, compressing depth-image and proprioceptive histories into a latent state that replaces hand-designed privileged information. Paper [4] explores **prediction without any environment-specific training**, delegating world modeling to pre-trained VLMs’ commonsense knowledge for zero-shot navigation. Together, these studies span physics-grounded, geometry-grounded, latent, and semantic designs, united by imagining outcomes before acting.

Table 3.1: Comparison of World-Model Approaches for Embodied AI

Paper Name	Methodology	Advantages	Limitations
[1]DexSim2Real ² : Explicit World Model for Articulated Object Dexterous Manipulation	Interactive perception + 3D AIGC digital twin construction; sampling-based MPC (iCEM) with eigengrasp dimensionality reduction for dexterous hands.	Demonstration-free and RL-free; high precision (avg. error $\approx 7^\circ$); supports multiple end effectors and unseen strategies such as tool use.	Requires observable state changes during interaction; 3D AIGC meshes may misalign for novel objects; minutes-scale trajectory search.
[2]GAF: Gaussian Action Field as a 4D Representation for Dynamic World Modeling	Extends 3DGS with learnable per-Gaussian motion; ViT-based feed-forward reconstruction; ICP-derived init action refined by action-vision-aligned diffusion.	Unifies reconstruction, prediction, and action; needs only sparse unposed RGB; +7.3% success over SOTA and real-time inference (<0.3 s).	Lacks semantic/task-level understanding; partial occlusions cause misaligned contacts; needs some real action data for refinement.

Paper Name	Methodology	Advantages	Limitations
[3]World Model-based Perception for Visual Legged Locomotion (WMP)	RSSM world model trained in simulation; policy on recurrent latent state via asymmetric actor-critic PPO; AMP-style natural-gait rewards.	Removes privileged teacher-student gap; near-teacher returns; best real-world traversal on Unitree A1 (Gap 85 cm, Climb 55 cm, Crawl 22 cm).	World model bounded by simulator fidelity; latency forces coarse model interval ($k = 5$); no real-data model training yet.
[4]WMNav: Integrating Vision-Language Models into World Models for Object Goal Navigation	PredictVLM scores directions into an online Curiosity Value Map; subtask decomposition; two-stage action proposer; all reasoning by prompted VLMs.	Zero-shot and training-free; +3.2%/+13.5% SR over SOTA on HM3D/MP3D; reduces risky exploration through prediction.	Dependent on proprietary VLM capability and cost; susceptible to hallucination; indoor object-goal setting only.

Model Performance and Evaluation: All four studies provide rigorous quantitative evaluation on real robots or standard benchmarks.

- Paper [1] achieved average manipulation errors of 7.2° (laptop) and 6.8° (cabinet) with the suction gripper versus $16.0^\circ/15.3^\circ$ for UMPNet, and world-model construction accuracy with joint-axis angle error below 9° across four object categories.
- Paper [2] reported a 60.4% average success rate on nine RLBench tasks, exceeding Diffusion Policy by 15.7%, Act3D by 7.3%, and ManiGaussian by 10.3%, alongside +11.54 dB PSNR reconstruction gains and 6–10/10 success on five real Franka tasks.

- Paper [3] attained near-teacher returns in simulation and the best real-world traversal performance reported on a Unitree A1: Gap 85 cm, Climb 55 cm, Tilt 28 cm, and Crawl 22 cm, terrains on which the privileged student baseline partially or completely fails.
- Paper [4] delivered state-of-the-art zero-shot navigation with 58.1% SR / 31.2% SPL on HM3D and 45.4% SR / 17.2% SPL on MP3D, and 72.2% SR on HM3D v0.2 in ablation configuration.

Despite these strong results, the evaluation settings differ substantially—single fixed-arm tabletop scenes, RLBench and Franka tasks, indoor quadruped courses, and Habitat photorealistic scans—so cross-paper numbers indicate domain-specific maturity rather than direct superiority of one representation over another.

Data and Environment Utilization: Data requirements diverge sharply across the four systems. Paper [1] needs only $K + 1$ real observation frames per object plus offline affordance training data, sourced either from simulated interaction (Where2Act on SAPIEN assets) or from large-scale egocentric human videos (VRB), the latter removing dependence on scarce interactable 3D assets. Paper [2] trains from RGB video frames alone for reconstruction—no depth, poses, or 3D ground truth—plus 20 demonstrations per RLBench task for action refinement. Paper [3] exploits massively parallel simulation (4096 robots in Isaac Gym) with terrain curriculum and domain randomization, requiring no real-world data at all. Paper [4] requires no training data whatsoever, relying entirely on frozen VLMs and the HM3D/MP3D evaluation scenes. This progression—from instance-specific reconstruction to zero-data prediction—highlights how world models can trade data collection against prior knowledge, whether physical, geometric, or semantic.

Scalability and Deployment Potential: Deployment characteristics vary significantly across the proposed systems:

- Paper [1]’s pipeline runs trajectory search in about 1.5–3 minutes per task on a desktop CPU/GPU, suitable for deliberate household manipulation but not for reactive control; its modular design extends naturally to new end effectors and tools.

- Paper [2] operates feed-forward on a single GPU with sub-0.3 s action prediction and parallel prediction–execution threads, making it genuinely real-time and deployable on real manipulators.
- Paper [3] runs fully on-board a Jetson NX without external computation, with the model interval $k = 5$ chosen explicitly to absorb 40 ms of sensing and inference latency—evidence of careful hardware–algorithm co-design.
- Paper [4]’s reliance on cloud VLM APIs introduces latency and cost per step, restricting current deployment to non-time-critical navigation, though the framework improves automatically as VLMs advance.

In real-world implementation, the trade-off between model fidelity, planning latency, and computational footprint must be balanced against task tempo: manipulation tolerates minutes of deliberation, locomotion demands 50 Hz reactivity, and navigation lies in between.

Interpretability and Trustworthiness: World models differ markedly in how inspectable their predictions are. Paper [1] is the most interpretable: its digital twin is a human-readable URDF with explicit joint types, axes, and meshes, and planned trajectories can be visualized in simulation before execution. Paper [2] renders photorealistic images of both current and predicted future scenes, allowing direct visual verification of what the model believes will happen. Paper [4] produces natural-language explanations for every directional score and subtask, and its feedback loop between plan and observation explicitly counteracts VLM hallucination. Paper [3], by contrast, encodes knowledge in a latent state; interpretability is indirect, through decoded depth predictions and t-SNE terrain clustering. This ordering suggests a tension between representation compactness and transparency that future world-model designs must negotiate.

Innovation and Research Contributions: Each paper introduces novel elements that advance the state-of-the-art:

- **Paper [1]** — First framework to combine interactive perception, 3D AIGC digital-twin reconstruction of multi-part articulated objects, and eigengrasp-reduced sampling-

based MPC for dexterous hands, with video-based affordance learning replacing scarce simulation assets.

- **Paper [2]** — Introduced the V-4D-A paradigm and the Gaussian Action Field, extracting actions from predicted Gaussian motion via ICP and refining them with actionable multiview RGB guidance.
- **Paper [3]** — First application of Dreamer-style RSSM world modeling to challenging vision-based legged locomotion with successful sim-to-real transfer, replacing the privileged learning paradigm.
- **Paper [4]** — Proposed the first fully modular VLM world model for ZSON, with the online Curiosity Value Map as quantitative predictive memory and hallucination-mitigating subtask feedback.

Together, these contributions indicate a shift toward embodied agents that construct, maintain, and exploit predictive models of their surroundings as a first-class component of the control loop.

Practical Relevance and Real-World Application: From a deployment perspective, Paper [1] targets assistive and household robotics where precise operation of cabinets, drawers, laptops, and lamps matters, and its tool-use extension previews service robots that improvise with available implements. Paper [2]’s real-time pipeline suits industrial and tabletop manipulation cells where sparse cameras are cheaper than calibrated depth rigs. Paper [3] directly advances inspection, delivery, and search-and-rescue quadrupeds that must negotiate stairs, gaps, and crawl spaces outdoors, as demonstrated in park deployments. Paper [4] addresses domestic assistance—finding objects in unseen homes—without any environment-specific setup. Collectively, these models cover deliberate manipulation, agile locomotion, and semantic navigation, demonstrating the versatility of the world-model paradigm across the full spectrum of embodied robotics.

Overall Comparative Insights: In synthesis, Paper [1] prioritizes precision through explicit physics, Paper [2] unification of prediction and action through 4D geometry, Paper [3] perceptual compression through latent dynamics, and Paper [4] zero-shot semantics through foundation models. Each contributes to a different dimension of world-model development:

- **Precision and Physical Grounding:** Papers [1]–[2] anchor predictions in explicit geometry and physics, enabling accurate contact-rich manipulation.
- **Sample Efficiency:** Papers [1] and [3] show that physics priors and simulated world-model training dramatically cut real-world data needs.
- **Real-Time Capability:** Papers [2]–[3] achieve on-board, low-latency inference; Paper [1] trades latency for precision; Paper [4] is bounded by VLM API speed.
- **Generality and Zero-Shot Ability:** Paper [4] generalizes to unseen scenes and categories without training, while Paper [1] generalizes to unseen actions such as tool use.

In conclusion, the four reviewed research papers collectively mark a significant evolution in embodied AI. The fusion of explicit digital twins, dynamic Gaussian representations,

latent recurrent dynamics, and VLM-based imagination has transformed robots from reactive controllers into agents capable of predicting, planning, and explaining their behavior. Future research will likely converge these paradigms—combining the precision of explicit models, the real-time efficiency of feed-forward 4D representations, the compression of latent states, and the semantic breadth of foundation models—to create unified, resilient world models for the next generation of embodied intelligent systems.

Chapter 4

Challenges

Despite significant progress in world-model-driven embodied AI, several technical, operational, and practical barriers still hinder effective deployment in unstructured real-world environments. The following summarizes the key challenges identified from the four reviewed studies.

Paper [1] (DexSim2Real² — Explicit World Model for Dexterous Manipulation): The framework infers kinematic structure by analyzing inter-frame differences, so it strictly requires observable state changes of every movable part during interactive perception; parts that cannot be moved or seen cannot be modeled. For novel or uncommon objects, existing 3D AIGC techniques may generate meshes that are imperfectly aligned with real-world observations, degrading digital-twin accuracy and hence manipulation precision, since manipulation error correlates positively with joint position and orientation errors of the model. Real objects also exhibit unmodeled dynamics—elastic deformation of laptop lids and rail gaps that turn ideal prismatic joints into multi-DOF joints—which the rigid simulation cannot capture. Finally, trajectory search takes 1.5–3 minutes per task, precluding reactive behavior.

Paper [2] (GAF — 4D Gaussian World Modeling): Although GAF unifies reconstruction, prediction, and action, its purely geometric world model lacks semantic and task-level understanding, limiting context-aware manipulation. Reconstruction errors under partial occlusion lead to misaligned contacts during interaction; without the diffusion-based refinement stage, executed init actions frequently fail, showing that visually plausible motion is not automatically physically feasible. The refinement stage reintroduces a dependence on real action demonstrations, and real-world failures concen-

trated in grasping (apple, door handle) reveal the absence of force feedback, suggesting the need for tactile sensing and richer gripper-state training data.

Paper [3] (WMP — World Model-based Perception for Locomotion): Because the RSSM is trained exclusively on simulated data, it cannot exceed the simulator’s fidelity; unmodeled real-world dynamics and never-seen obstacle shapes manifest as prediction discrepancies (e.g., the Crawl obstacle’s shape differs in predicted depth images). Real-world latency of depth acquisition and on-board inference (≈ 40 ms) forces a coarser world-model interval ($k = 5$), sacrificing the higher returns that faster model updates achieve in simulation. Backpropagating gradients through long RSSM training segments is inefficient, restricting usable history length to seconds. Training the world model with mixed simulated and real data, and incorporating additional sensing modalities such as touch, remain open future work.

Paper [4] (WMNav — VLM World Models for Navigation): The framework inherits the weaknesses of its foundation models: VLM hallucination can inject erroneous predictions into the Curiosity Value Map, and although subtask feedback and quantitative scoring mitigate this, they cannot eliminate it. Performance scales strongly with VLM capability—open-source Qwen2.5-VL models trail proprietary Gemini models by over 10% SR—creating cost, latency, and accessibility dependencies on commercial APIs. Each navigation step requires multiple VLM calls, limiting real-time operation. The evaluation is confined to indoor object-goal tasks in Habitat scenes; extension to dynamic environments, outdoor settings, and instruction-following tasks is untested.

In summary, world-model-based embodied AI faces challenges in model fidelity, sim-to-real alignment, semantic grounding, latency, and dependence on foundation-model quality. Addressing these issues requires hybrid training on simulated and real data, integration of tactile and force sensing, semantic priors within geometric world models, hallucination-resistant prompting and verification mechanisms, and efficient on-board inference. Progress along these directions will be essential to develop reliable, transparent, and broadly deployable world models for next-generation embodied intelligent systems.

Chapter 5

Conclusion

The integration of **world models** into **embodied AI** marks a major milestone in the evolution of intelligent robotics. Purely reactive policies, which map current observations directly to actions, often fail under the partial observability, complex dynamics, and open-endedness of the physical world. World-model-driven agents overcome these limitations by maintaining an internal, predictive representation of their environment—imagining the consequences of candidate actions before execution—thereby enabling precise planning, efficient exploration, and robust sim-to-real transfer across manipulation, locomotion, and navigation.

This study reviewed four key research contributions illustrating the diverse realizations of world models in embodied AI. **Paper [1]** (DexSim2Real²) constructed an *explicit* digital-twin world model of unseen articulated objects through interactive perception and 3D AIGC reconstruction, enabling sampling-based MPC with eigengrasp-reduced dexterous hands to achieve goal-conditioned manipulation errors of only a few degrees—without demonstrations or reinforcement learning—and generalizing to advanced skills such as tool use. **Paper [2]** (GAF) introduced a 4D Gaussian Action Field that unifies scene reconstruction, future prediction, and action extraction in a single feed-forward framework, improving average manipulation success by 7.3% over state-of-the-art baselines while running in real time from sparse unposed RGB views. **Paper [3]** (WMP) demonstrated that an RSSM world model trained entirely in simulation can serve as the perceptual backbone for visual legged locomotion, achieving the best reported traversal performance on a Unitree A1 quadruped and eliminating the teacher–student gap of privileged learning. Finally, **Paper [4]** (WMNav) integrated Vision-Language Models into a modular world model with an online Curiosity Value Map, attaining state-of-the-art zero-shot object goal navigation on HM3D and MP3D without any task-specific training.

Taken together, these works trace a coherent design space—explicit physics-based, geometry-embedded, latent, and semantic world models—each optimizing a different balance among precision, data efficiency, latency, interpretability, and generality. The comparative analysis indicates that no single representation dominates; rather, the appropriate world model depends on the tempo and structure of the task at hand. Future research is expected to converge these paradigms: enriching explicit and Gaussian world models with semantic priors from foundation models, training latent models on mixtures of simulated and real experience, incorporating tactile and force sensing, and scaling from single-object digital twins to scene-level world models for long-horizon mobile manipulation. Such convergence promises embodied agents that perceive, imagine, plan, and act with human-like foresight, bringing general-purpose robotic intelligence closer to practical reality.

References

- [1] T. Jiang, Y. Guan, L. Ma, J. Xu, J. Meng, W. Chen, Z. Zeng, L. Li, D. Wu, and R. Chen, “DexSim2Real²: Building Explicit World Model for Precise Articulated Object Dexterous Manipulation,” *IEEE Transactions on Robotics*, 2025.
- [2] Y. Chai, L. Deng, R. Shao, J. Zhang, K. Lv, L. Xing, X. Li, H. Zhang, and Y. Liu, “GAF: Gaussian Action Field as a 4D Representation for Dynamic World Modeling in Robotic Manipulation,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [3] H. Lai, J. Cao, J. Xu, H. Wu, Y. Lin, T. Kong, Y. Yu, and W. Zhang, “World Model-based Perception for Visual Legged Locomotion,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11531–11537.
- [4] D. Nie, X. Guo, Y. Duan, R. Zhang, and L. Chen, “WMNav: Integrating Vision-Language Models into World Models for Object Goal Navigation,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 2392–2399.